

■ **Power Mode:** Best Performance

This setup represents a realistic yet high-performance environment for creators and power users. The combination of quad-channel DDR5-6400 memory and PCIe Gen 5 NVMe storage ensures that the CPU isn't starved for bandwidth, while the RTX 5090 provides ample headroom for GPU-accelerated workloads.

PERFORMANCE

Cinebench R23

Cinebench R23 is a staple in the CPU benchmarking world, simulating real-world CPU rendering using the Cinema 4D engine. Its multi-core score shows raw throughput, while the single-core score reflects responsiveness and IPC.

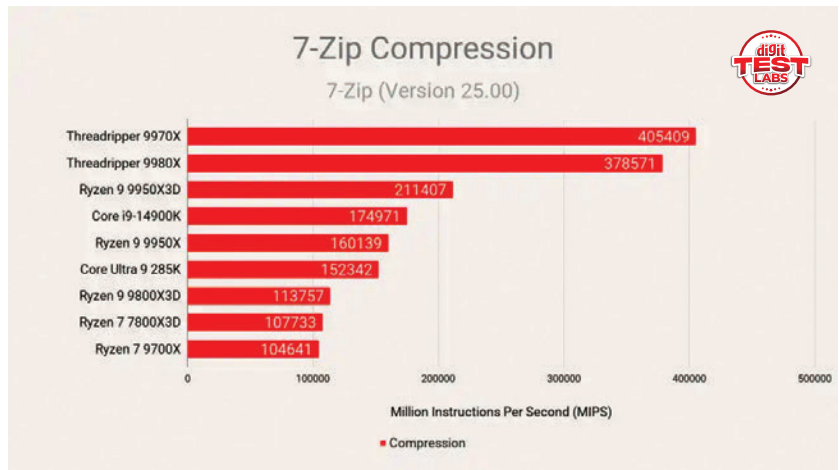
The 9970X achieved 76,392 points in the multi-core test, trailing the 9980X by about 25 percent but coming in comfortably ahead of the 9950X3D. In single-core performance, it scored 2,162 points, which is virtually identical to the 9980X.

These results reflect what the 9970X is optimised for: strong all-core rendering performance with minimal compromise on per-thread responsiveness. For artists working with Cinema 4D, Redshift, or similar tools, this CPU offers excellent balance between live viewport interactivity and final render throughput.

Blender

Blender's Cycles engine is a real-world renderer that's fully open source, popular in game design, animation, and indie filmmaking. Its benchmark includes three scenes: Monster (GPU-heavy), Junkshop (hybrid), and Classroom (CPU-heavy).

In CPU-only mode, the 9970X rendered the Classroom scene with 531.302 samples/second, Junkshop with 373.357 samples/second, and Monster in 283.349 samples/second. The 9980X was faster on all three counts, but not drastically so, most notably in the Junkshop scene where the difference was just 12 seconds.



These scores suggest that the 9970X is more than capable of serving as a primary Blender workstation. Unless you are pushing multi-layer compositing across several scenes simultaneously, the 9970X offers excellent frame-to-frame responsiveness and competitive final render times.

Procyon AI Inference

The Procyon AI benchmark uses ONNX models like ResNet50, MobileNet V3, and BERT to simulate real-world AI workloads running on the CPU. These workloads include image classification, natural language inference, and feature extraction which are all key components in local productivity and enterprise AI applications.

The Threadripper 9970X scored 288 on the CPU inference suite, besting the 9980X slightly and outperforming most mainstream desktop CPUs. In practical terms, this means AI-powered applications like Photoshop's Neural Filters or After Effects' Scene Edit Detection run with greater fluidity and reduced latency.

While the 9970X lacks a dedicated NPU (unlike mobile processors in AMD's Ryzen AI family), its 32 Zen 5 cores provide enough throughput to handle low-latency model inference, automation scripts, and integrated ML tools for creators and researchers alike.

THERMALS

Thermally, the Ryzen Threadripper 9970X runs slightly warmer than its

flagship sibling, despite featuring fewer cores. Under full load, the 9970X reached peak temperatures of around 75°C, with idle readings near 39°C. In contrast, the 64-core 9980X maintained a cooler profile, topping out at 65.6°C under load and idling at approx 36°C.

VERDICT

The AMD Ryzen Threadripper 9970X stands out as the most balanced offering in the new Threadripper 9000 lineup. With its 32-core, 64-thread configuration, it brings together high sustained performance, excellent thermal behaviour, and architectural efficiency in a way that feels purpose-built for demanding professional workloads. From large-scale 3D renders and simulations to video encoding, the 9970X handles it all with minimal fuss and maximum consistency.

—Mithun Mohandas

SPECIFICATIONS

Architecture: Zen 5 | Cores/Threads: 32/64 | Base Clock: 4.0GHz | Boost Clock: 5.4GHz | L2/L3 Cache: 32MB/128MB | PCIe 5.0 Lanes: 80 | Memory Support: DDR5-6400 | ECC Support | Overclocking Unlocked | TDP: 350 W | Socket: sTR5 | Boost Algorithm: Precision Boost 2 | Max Temperature (TJMax): 95°C

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